Loan Approval Prediction

AI for Engineers (UCS321) — MST Mini Project, Problem 5

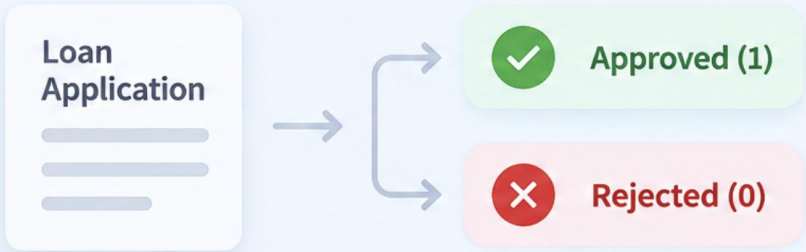
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Problem Statement

The project aims to develop a predictive model to determine whether a bank approves or rejects a loan application based on applicant features.

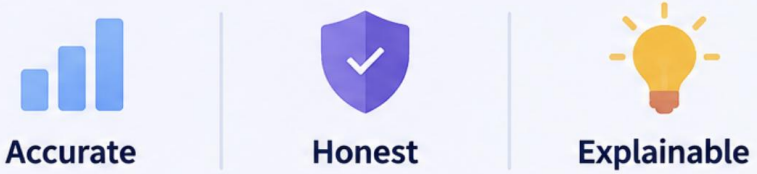
Binary Classification Task

The problem is structured as a binary classification, where the model must predict one of two outcomes: Approved (1) or Rejected (0).



Model Objectives

The primary goal is to build a model that is not only accurate but also honest and explainable, enabling the identification of the key applicant features that drive the loan decision.



Dataset Overview

Data Profile

Dataset Size
4,269
loan applications

Features
13
columns of applicant data

Data Quality
The dataset is clean, with no missing values or duplicate rows.

Key Features

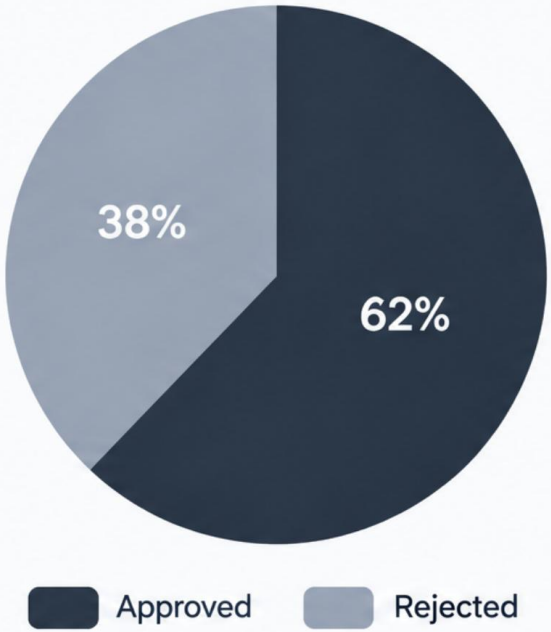
Financial
Income, loan amount, loan term

Credit History
CIBIL score

Demographic
Education, self-employment

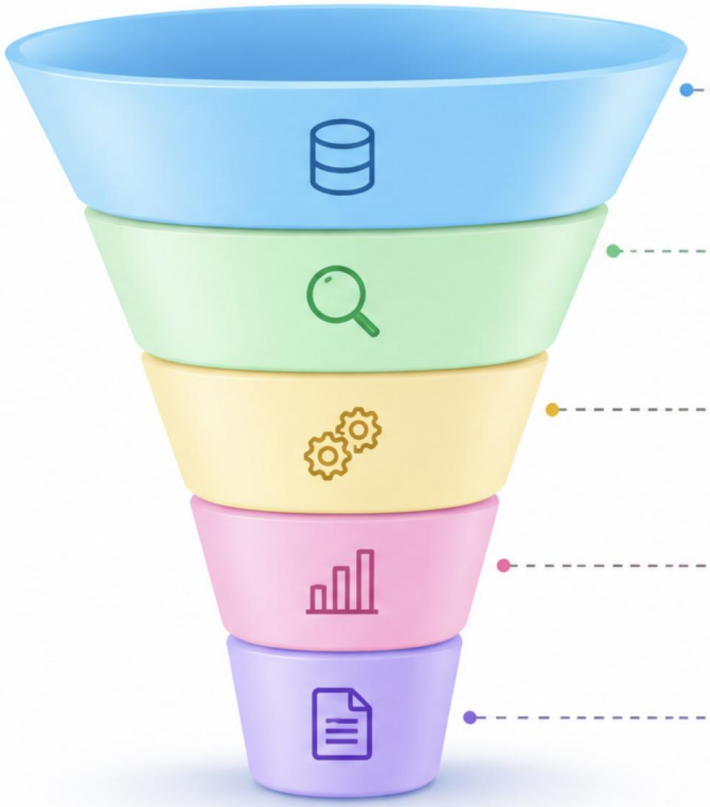
Assets
4 asset-value columns

Target Variable: loan_status



Methodology: Project Pipeline

The project followed a structured machine learning pipeline, progressing from data preparation and analysis to model training, evaluation, and interpretation.



01 **Data Preparation**
Raw Data, Strip & Clean

02 **Exploratory Data Analysis (EDA) & Feature Engineering**

03 **Data Processing**
Encode, Split 80/20, Scale

04 **Modeling**
Train 3 Models & Tune

05 **Evaluation & Interpretation**
Evaluate, Importance & Ablation

Exploratory Data Analysis (EDA)

Initial analysis revealed that the CIBIL (credit) score is the most dominant predictor of loan approval, far outweighing other features.

Weak Predictors

These features showeed minimal impact on loan approval rates.

Education

Approval rates showed minimal difference between graduates and non-graduates.

Self-Employment

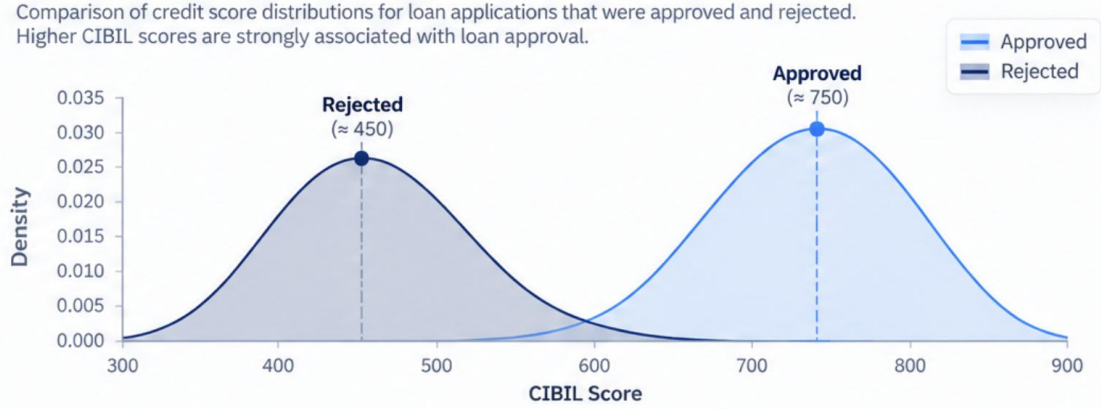
The approval rate was nearly identical for both self-employed and non-self-employed applicants.

Dominant Predictor: CIBIL Score

The CIBIL score distribution shows a clear and significant separation between approved and rejected applications. Approved loans are strongly correlated with high scores, while rejected loans are associated with low scores.

CIBIL Score Distributions: Approved vs. Rejected

Comparison of credit score distributions for loan applications that were approved and rejected. Higher CIBIL scores are strongly associated with loan approval.



Approved
Mean CIBIL Score: ~ 750 (higher creditworthiness)

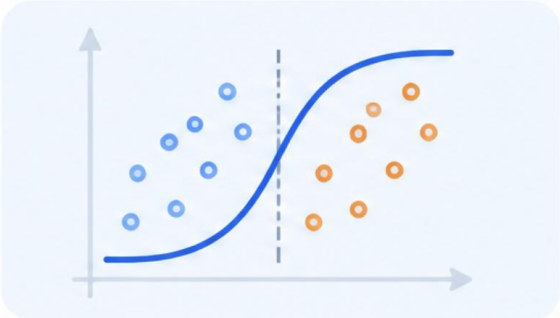
Rejected
Mean CIBIL Score: ~ 450 (lower creditworthiness)

Key Insight
Higher CIBIL scores are strongly associated with loan approval.

Takeaway
Improving credit score increases the likelihood of loan approval.

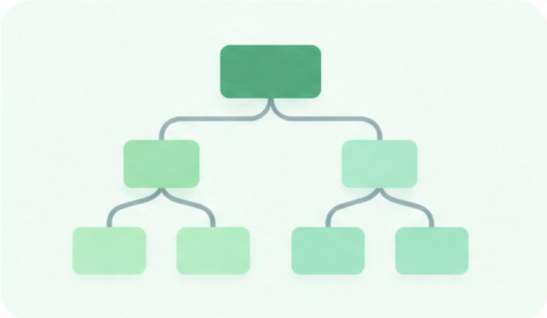
Model Selection & Configuration

A suite of three models was chosen to balance interpretability and performance.
All models were developed within a standardized pipeline to ensure consistency and reproducibility.



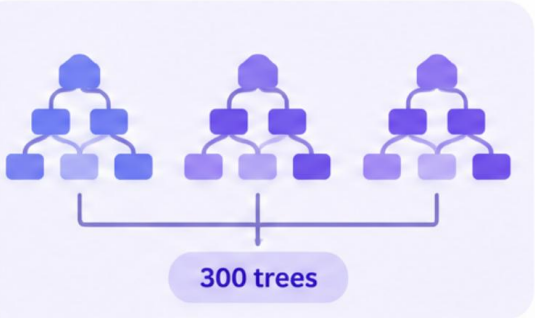
1 Logistic Regression

Utilized as a simple, highly interpretable baseline to establish a performance benchmark.



2 Decision Tree (max_depth=4)

Chosen for its ability to capture non-linear decision boundaries while remaining visually explainable.

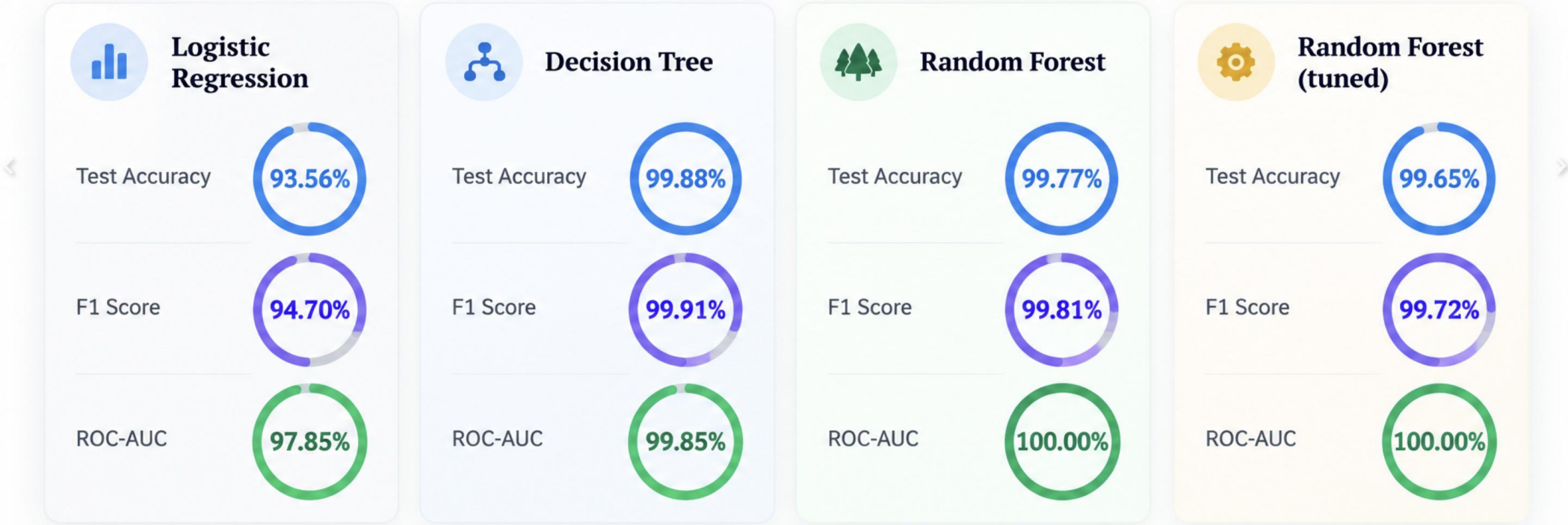


3 Random Forest (300 trees)

A powerful ensemble model, tuned with GridSearchCV, to maximize predictive accuracy.

Model Performance Results

All models were evaluated on the held-out test set. The Decision Tree and Random Forest models achieved exceptionally high performance across all metrics, indicating a highly predictable pattern in the data.



Feature Importance & Ablation Study

The near-perfect accuracy (>95%) raised an 'honesty flag,' prompting a deeper investigation into the features driving the model's decisions.

CIBIL Score Dominance

Analysis confirmed that `CIBIL_score` alone is the primary driver of outcomes. A simple rule-based model using only a CIBIL score threshold achieved 95.4% accuracy, highlighting its overwhelming predictive power.



Rule-based model
(CIBIL score only)

Ablation Study Impact

To quantify this effect, an ablation study was conducted. Removing the `CIBIL_score` feature caused the tuned Random Forest's F1 score to collapse from 0.997 to 0.662, a catastrophic 33.7% performance decrease.



33.7%
Performance decrease

Conclusion & Future Work

Key Findings

Best Model

The Decision Tree and tuned Random Forest models performed exceptionally well, achieving near-perfect F1 scores (≈0.997–0.999).

Dominant Feature

`cibil_score` is the unequivocal driver of loan approvals, confirmed by multiple importance methods and an ablation study.

Dataset Nature

The dataset is likely synthetic or simplified due to the near-deterministic relationship between credit score and loan status.

Limitations & Next Steps

Limitations

The analysis lacks a formal fairness audit and may be subject to proxy bias via education or employment features.

Future Work

Proposed enhancements include using XGBoost, exploring real-world bank data, applying SMOTE for data augmentation, and conducting a formal fairness audit.